Dessin D'enfants

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Outline

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We do not know any other explicit examples of elements of the absolute Galois group.

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Hence, it is nearly impossible for us to manually investigate the other elements of Γ, with such a high level of unpredictability, unlike the two already-known automorphisms.

We have to rely on something else.

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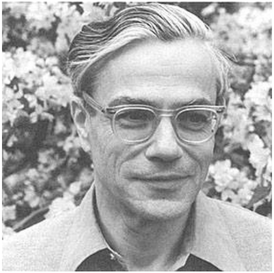

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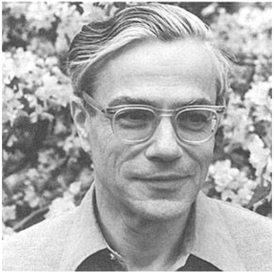

Analogously, instead of studying Γ itself, Grothendieck decided to interpret the action of Γ on “Dessin D'enfants”.

Definition of Dessin

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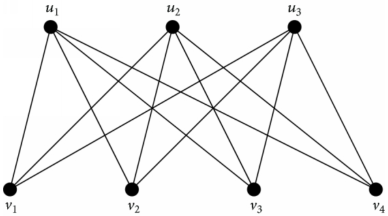
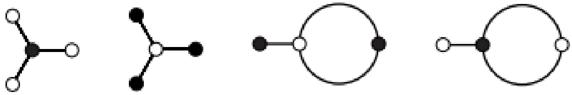
Definition: (Bigraph)

A bigraph is a connected bipartite graph with a fixed two coloring into nonempty sets of black & white vertices. (i.e. no adjacent vertices share the same color.)



Definition of Dessin

A <u>Dessin D'enfant</u> is a pair (X, D) where X is a compact oriented topological surface and D is a finite bipartite graph embedded in X such that $X \setminus D$ is a union of open disks (i.e. the faces of Dessin).



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- Dessin connects algebraic geometry and Galois theory mainly through "Belyi's Theorem".
- Belyi's theorem is derived from certain maps between algebraic curves.
    - These algebraic curves are defined over the algebraic numbers.
- Grothendieck showed that:

$$\text{Dessins} \quad \Longleftrightarrow \quad \text{maps derived from Belyi's theorem}$$

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(The white vertices is a combinatorial tool to ensure bipartiteness)

Belyi Map & Dessin

Let X be a compact Riemann surface, we will take Chow's theorem for granted, which asserts that "A compact Riemann surface is equivalent to an algebraic curve over $\mathbb{C}$".

Theorem: (Belyi's Theorem)

- X defined over $\overline{\mathbb{Q}}$ if and only if $\beta : X \to \hat{\mathbb{C}} = \mathbb{C} \cup \{\infty\}$ is a ramified covering map with ramification points $\{0, 1, \infty\}$.
- (i.e. β is the "Belyi map" - the definition is a consequence of the above theorem.)

Defintion: (Belyi Pair)

- (X, β): where X is a compact Riemann surface, β is the Belyi map.

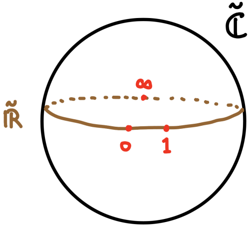
Belyi Map & Dessin

The LHS of BT suggests that our Riemann Surface / Algebraic Curve imposed with this algebraic structure, essentially has a topological correspondence.

The Belyi pair encodes the topological data associated with the covering.

Arise of Dessin from Belyi Map

Consider the Riemann surface $X = \tilde{\mathbb{C}}$.



We take the Belyi map and "lift up" the real projective line up to X. (i.e. we lift up the three ramification points and the segment connecting them.)

$$\overline{\beta^{-1}(0,1) \cup \beta^{-1}(1,\infty) \cup \beta^{-1}(\infty,0)}$$

$$=$$

triangulation of X with vertices $\beta^{-1}(0),\ \beta^{-1}(1),\ \beta^{-1}(\infty)$

Arise of Dessin from Belyi Map

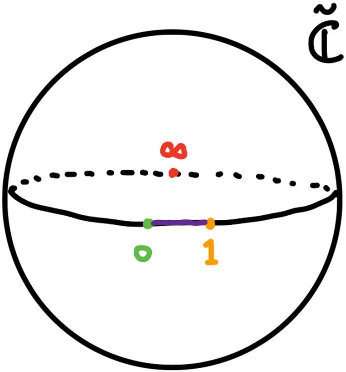
Hence, the fibre of ramification points are vertices, and the preimage of what's in between of the ramification points becomes the faces in triangulation. The table below is a summarized comparison of this "lift-up":

On $\widehat{\mathbb{C}}$	**On** X
One triangle	Many preimage triangles
Three branch points	Many vertices (fibers of 0, 1, ∞)
One face	Many faces covering X

The triangulation encodes all the topological data. However, to construct Dessin, we only need part of those data: $\beta^{-1}(0,1)$ with $\beta^{-1}(0)$ and $\beta^{-1}(1)$ coloured.

Arise of Dessin from Belyi Map

Hence, we have a bipartite graph embedded on the surface $\hat{\mathbb{C}}$



Arise of Dessin from Belyi Map

Proposition: (Construction of Dessin)

- Let β be a Belyi map for some X.
- The bigraph with black vertices $\beta^{-1}(0)$, white vertices $\beta^{-1}(1)$ and edges $\beta^{-1}(0,1)$, is a dessin embbed in X.

Example

Let $X = \hat{C}$ and $\beta(x) = x^2$, then let's consider the behaviour of the preimage of $0,\ 1$.

$\beta^{-1}(0) = 0$, with multiplicity 2, then it is a black vertex of valency 2.

$\beta^{-1}(1) = \pm 1$, each with multiplicity 1, then they are white vertices of valency 1.

Hence, the corresponding Dessin is:

Galois, Dessin, Belyi

Theorem: The isomorphism classes of Dessins are equivalent to the isomorphism classes of Belyi maps.

We take this for granted - but it is essentially saying that: to study Dessin, it suffices to study the Belyi map.

Galois, Dessin, Belyi

We first note that the field of algebraics $\overline{\mathbb{Q}} = \bigcup K$, where $K \supseteq Q$ are all finite Galois extension of $\mathbb{Q}$.

Each $g \in \Gamma$ is determined by its restriction to K : g_K.

Each K is preserved by the action of Γ, then $g_K \in \mathsf{Gal}(K/\mathbb{Q})$

Galois, Dessin, Belyi

Consider what we have so far:

$$\begin{aligned} & X \text{ compact Riemann surface over } \overline{\mathbb{Q}} \\ \iff & \text{ algebraic curves over } \overline{\mathbb{Q}} \\ \iff & \text{ zero locus of } F(x,y) \in \overline{\mathbb{Q}}[x,y] \end{aligned}$$

$$\implies \exists \sigma \in \mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}) \text{ acts on } X \text{ by acting on the coefficient of } F.$$

The takeaway is, this F what defines our X over $\overline{\mathbb{Q}}$ and our Belyi map is a rational function $\beta : X \to \mathbb{P}^1(\mathbb{C})$, and because Belyi maps determine dessins, so the effect of a nontrivial Galois action on the Belyi map will lead to a different Dessin with different combinaotiral data. We will gain insight of the absolute Galois group by distinguish such a difference.

Galois, Dessin, Belyi

$$\begin{tikzcd}
\mathcal{D} \arrow[rr, dashed] \arrow[d] & & \mathcal{D}^{\sigma} \\
(X, \beta) \arrow[rr, "\sigma"] & & (X^{\sigma}, \beta^{\sigma}) \arrow[u]
\end{tikzcd}$$

This is a commutative diagram that illusrates how the original Dessin is related to the new Dessin after the Galois action. We first note that Dessin defines the Beyli pair, and the Beyli pair gets mapped to the new Beyly pair under the action of σ, which associates to the new Dessin corresponding to this action.

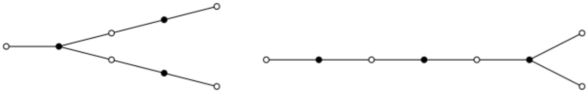
Galois, Dessin, Belyi

Theorem: (Grothendieck)

- Γ acts faithfully on X by acting on the coefficient of F.
- Definition: (faithfull) $\forall \sigma \neq \tau \in \Gamma,\ \sigma(x) \neq \tau(x)$ for some algebraic x.

Very Importantly: There is indeed an action from Γ on Dessin. However, the action itself is also not easy to understand, but by how simple dessins are - the difficulty is significantly dropped by analyzing the differences in their combinatorial data.

A Nontrivial Galois Action on a Dessin



The Belyi map is:

$$f(z) = z^3 \left(z^2 - 2z + \frac{34}{7} \pm \frac{6\sqrt{21}}{7} \right)^2$$

An element $\sigma \in \mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ that sends $\sqrt{21} \mapsto -\sqrt{21}$ acts on the coefficients of the Belyi map and sends the dessin on the left to the one on the right.

Thanks!